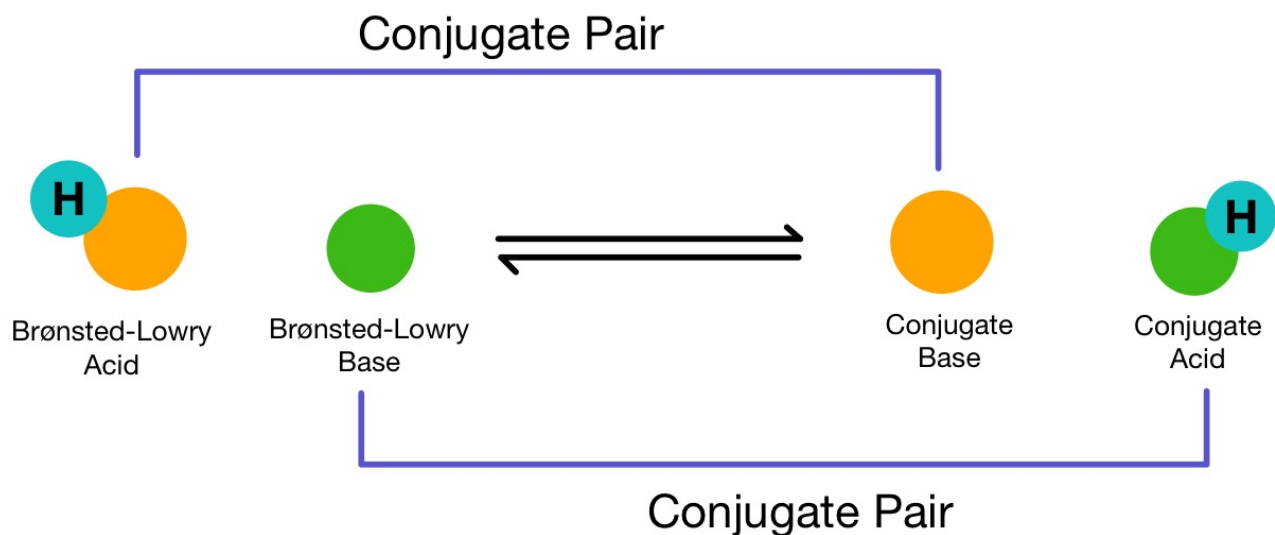


Chapters 17: Acid and Base



Dr. Wang

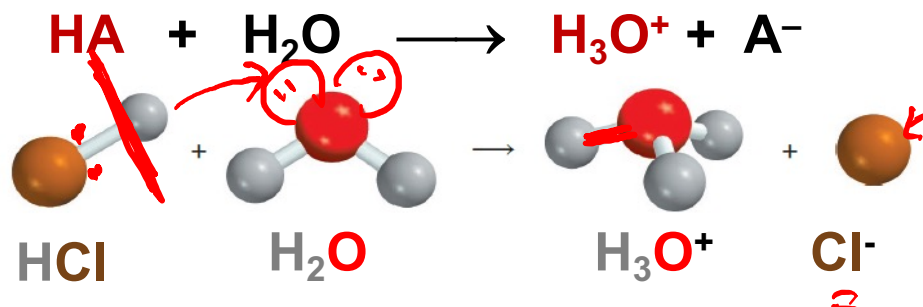
Learning Objectives

1. Understand the Arrhenius definition of acids and bases, and identify the products of
 - a. Ionization of an Arrhenius acid
 - b. Dissociation of an Arrhenius base
2. Describe the formation of hydronium ion from an aqueous acid
3. Understand the Bronsted-Lowry definition of acids and bases, and identify proton donor, proton acceptor, and conjugate acid-base pairs
4. Identify amphoteric species in a chemical reaction

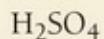
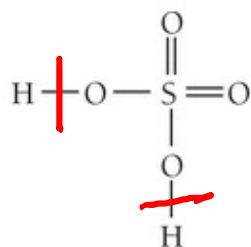
Arrhenius Acid-Base Theory

Arrhenius Acid:

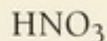
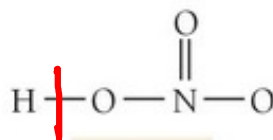
- A substance that can ionize into hydronium ion, $\text{H}_3\text{O}^+/\text{H}^+$, in water



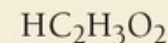
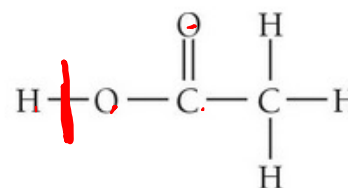
- Bond polarity: polar X-H bond can be broken to generate H^+



Sulfuric acid



Nitric acid

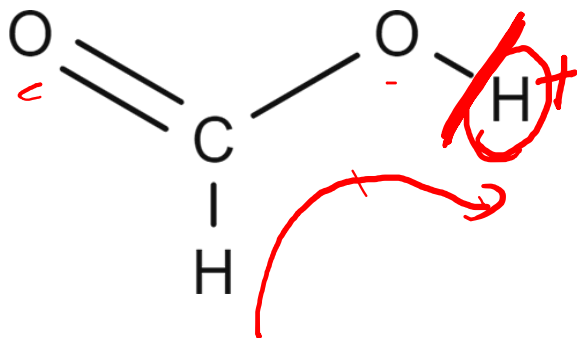


Acetic acid



Your turn!

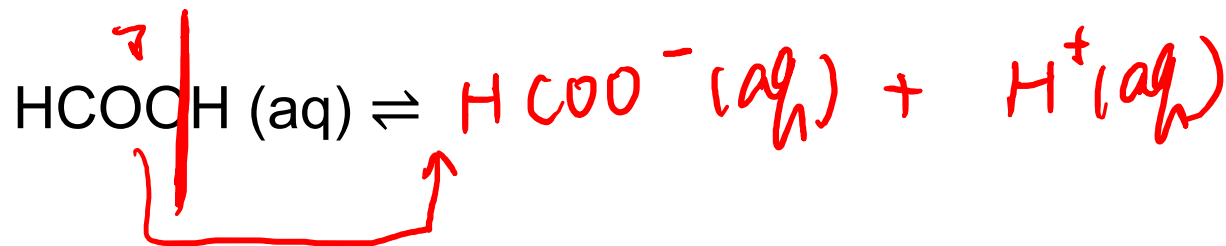
1. Below is the structural formula for formic acid. Which H atom will be ionized when formic acid is dissolved in water?



- A. H on the left
- ☒ B. H on the right
- C. They will both be ionized



2. Write out the ionization reaction for formic acid.



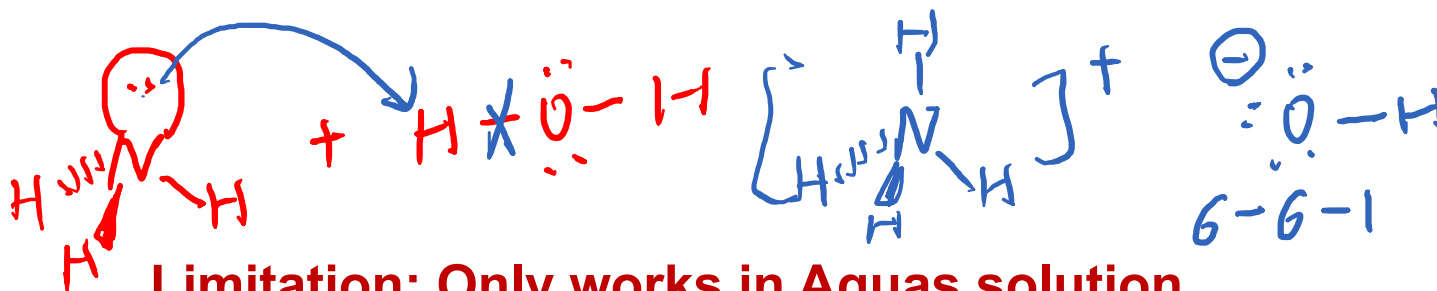
Arrhenius Acid-Base Theory

Arrhenius Base:

- A substance that can dissociate hydroxide ion, OH^- , in water



- Ionic Bases:** Metal Hydroxides and oxides
 - NaOH , Na_2O , Ca(OH)_2
- Molecular Bases:** Molecules



Announcement 4/29

Assignments that are due soon

- Pre-Class Assignment Week 13-F

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

1st Chem RePS or SI worksheets for Unit 4 will be closed on 11:59 PM 4/29

Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

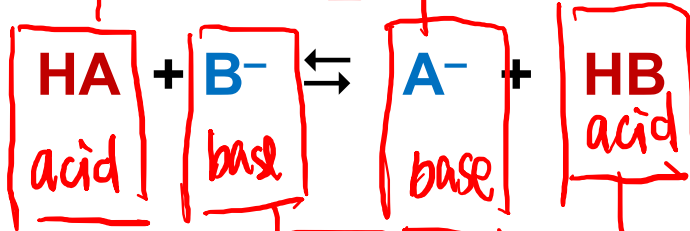
SI Session

Saige Douglass – Wed 5:30 – 7:30 PM at Hashinger 412

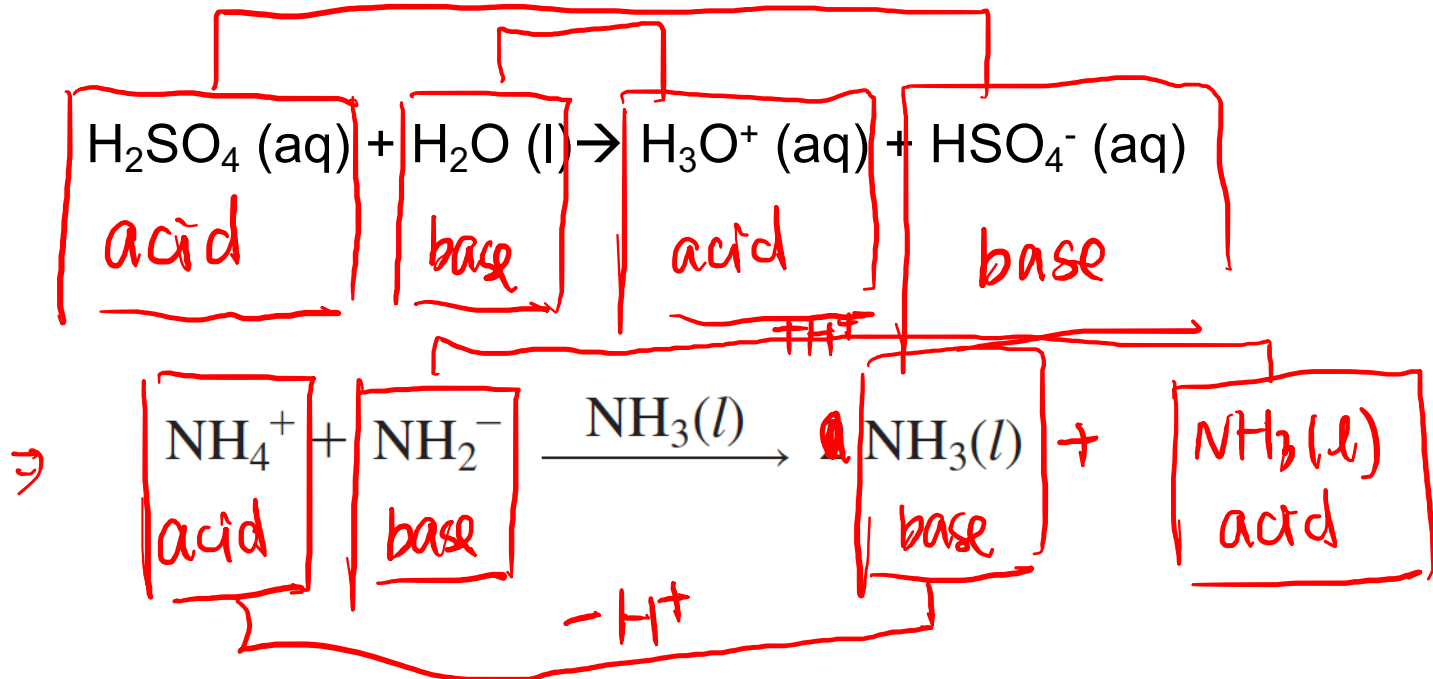
Paige Kwan - Thu 5 - 7 PM at KC134

Brønsted–Lowry Acid-Base Theory

- **Brønsted Acid:** A substance that tends to lose a H^+ (HA)
- **Brønsted Base:** A substance that tends to accept a H^+ (A^-)

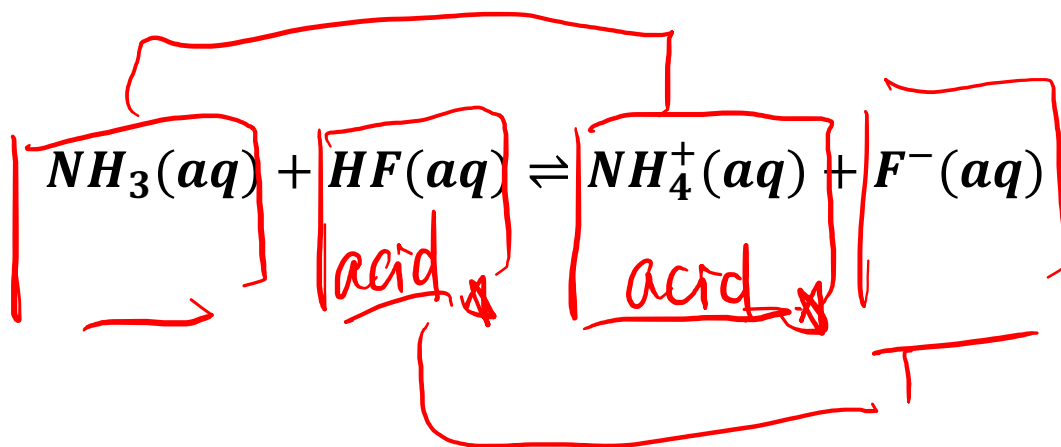


- **Conjugate Brønsted Acid-Base pair:** different by exactly 1 H^+
- In a Brønsted Acid-Base reaction:



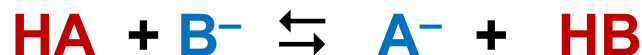
Your turn!

1. Identify the conjugate acid-base pairs in the following acid-base reactions:



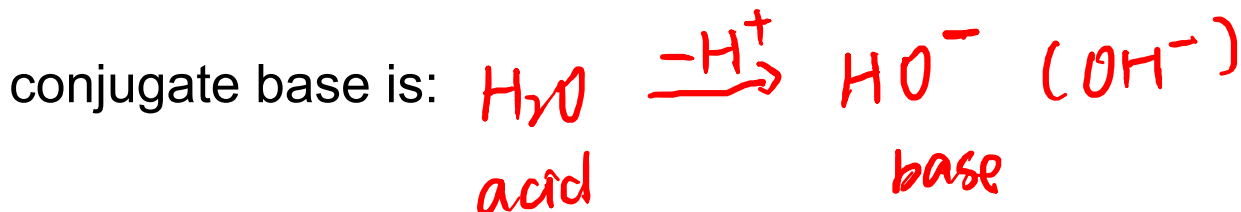
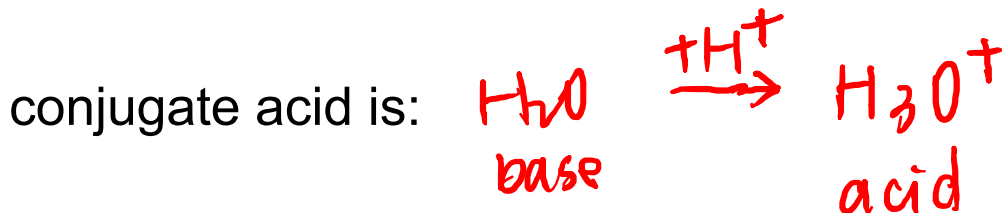
Brønsted–Lowry Acid-Base Theory

- **Brønsted Acid**: A substance that tends to lose a H^+
- **Brønsted Base**: A substance that tends to accept a H^+



- Amphoteric $\equiv$ reacts with both acids and bases.

Water (H_2O) is amphoteric

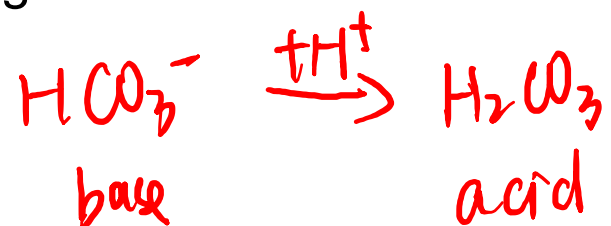


Your turn!

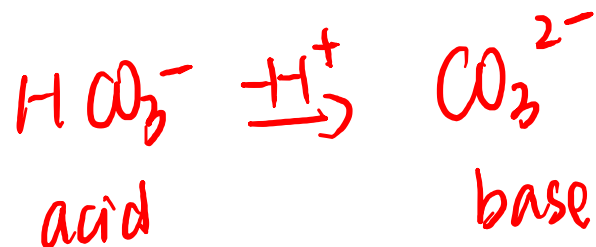
1. Identify the conjugate acid and conjugate base for each of the following species:



HCO_3^- 's conjugate acid is:

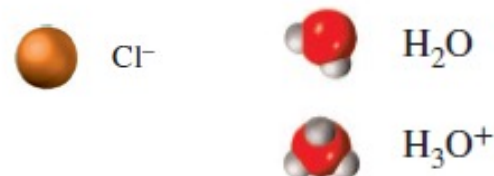
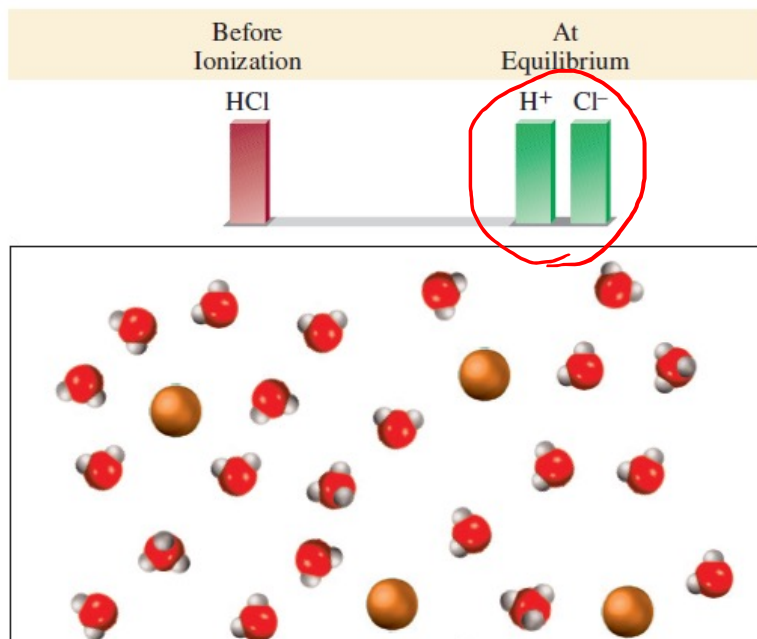
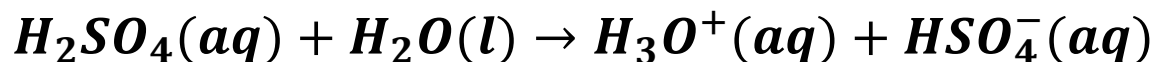
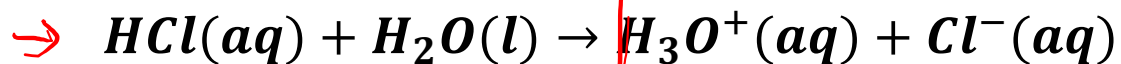


HCO_3^- 's conjugate base is:



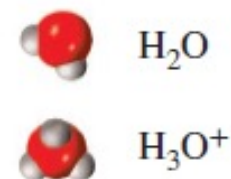
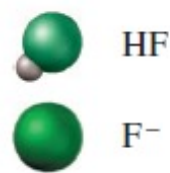
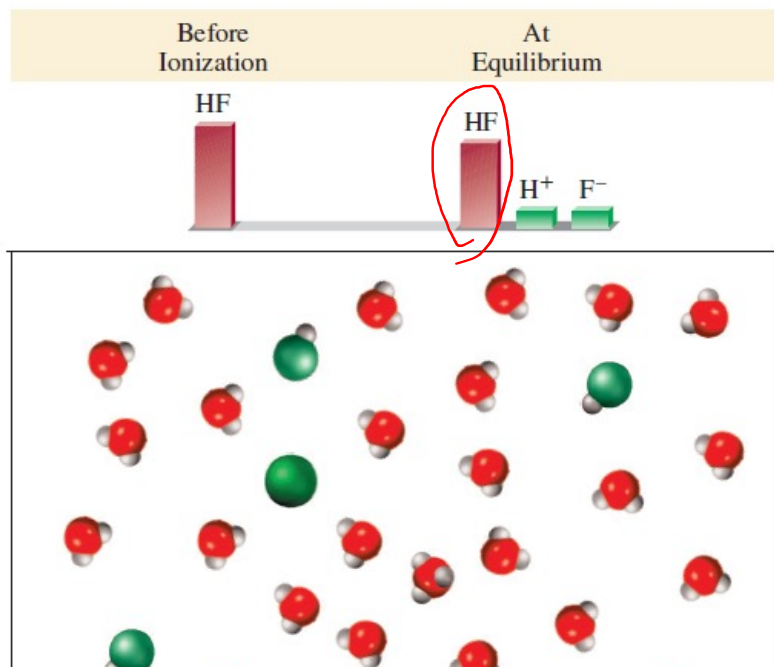
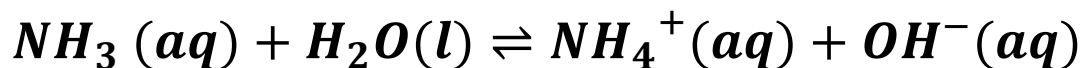
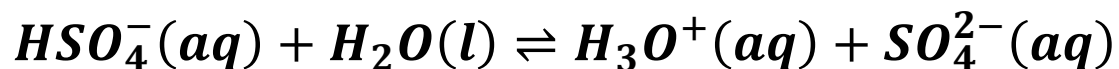
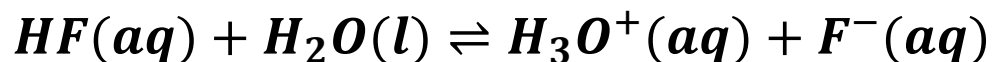
Strengths of Acids and Bases

- Strong acid/base ionize/dissociate completely (~100%) and at the equilibrium, only conjugated base / acid exists.

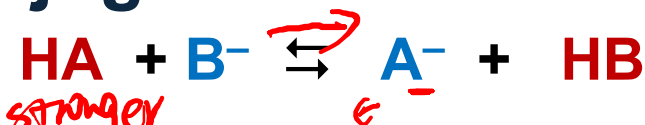


Strengths of Acids and Bases

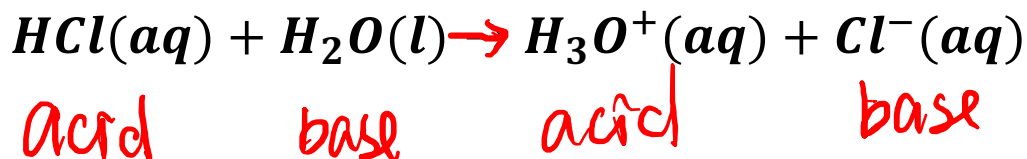
- Weak acid/base ionize/dissociate incompletely (< 100%) and at the equilibrium, both reactants and products exist.



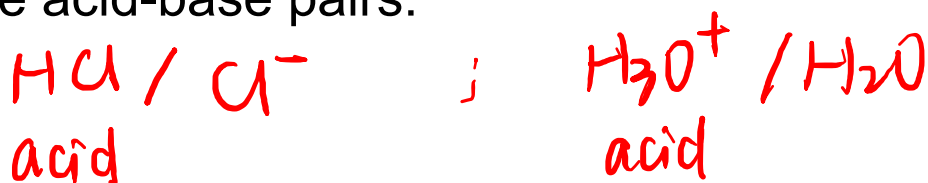
Conjugate Acid-Base Pairs



- strong acid and base tend to completely dissociate/ionize.
- The stronger an acid is, the weaker its conjugate base will be.
- The stronger a base is, the weaker its conjugate acid will be.
- At the equilibrium, weaker acid and base is preferred.



Conjugate acid-base pairs:

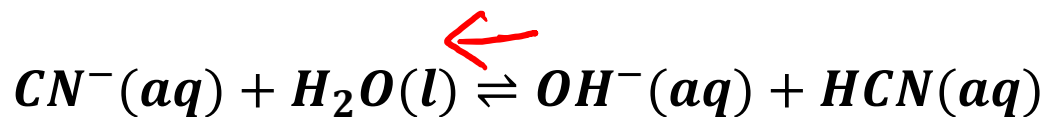


Compared to HCl, the weaker acid is H₃O⁺

Compared to H₂O, the weaker base is Cl⁻

Your turn!

Consider the following reaction:



base *base*
Which chemical specie is the weaker base?

$$K = 2.04 \times 10^{-5} \ll 1$$

Hint: Identify the conjugate acid-base pairs first.

☒ A. $\text{CN}^-(aq)$

B. $\text{H}_2\text{O}(l)$

C. $\text{OH}^-(aq)$

D. $\text{HCN}(aq)$



Amphoteric Water



Table 15.2 Relative Strengths of Conjugate Acid-Base Pairs

	Acid	Conjugate Base	
Acid strength increases ↑	Strong acids		
	HClO ₄ (perchloric acid)	ClO ₄ ⁻ (perchlorate ion)	
	HI (hydroiodic acid)	I ⁻ (iodide ion)	
	HBr (hydrobromic acid)	Br ⁻ (bromide ion)	
	HCl (hydrochloric acid)	Cl ⁻ (chloride ion)	
	→ H ₂ SO ₄ (sulfuric acid)	HSO ₄ ⁻ (hydrogen sulfate ion)	
	HNO ₃ (nitric acid)	NO ₃ ⁻ (nitrate ion)	
	→ H ₃ O ⁺ (hydronium ion)	H ₂ O (water)	
	Weak acids		
	HSO ₄ ⁻ (hydrogen sulfate ion)	SO ₄ ²⁻ (sulfate ion)	
	HF (hydrofluoric acid)	F ⁻ (fluoride ion)	
	HNO ₂ (nitrous acid)	NO ₂ ⁻ (nitrite ion)	
	HCOOH (formic acid)	HCOO ⁻ (formate ion)	
	CH ₃ COOH (acetic acid)	CH ₃ COO ⁻ (acetate ion)	
	NH ₄ ⁺ (ammonium ion)	NH ₃ (ammonia)	
	HCN (hydrocyanic acid)	CN ⁻ (cyanide ion)	
	H ₂ O (water)	OH ⁻ (hydroxide ion)	
	NH ₃ (ammonia)	NH ₂ ⁻ (amide ion)	
			Base strength increases ↓

Your turn!

HSO_4^- and HPO_4^{2-} are both amphoteric. Knowing that HSO_4^- is a stronger acid than HPO_4^{2-} . Write out the chemical reaction of a mixture with HSO_4^- (aq) and HPO_4^{2-} (aq). Hint: The stronger acid have larger tendency to give out the H^+ .

